

Battery Charge Threshold Setup Guide

Ubuntu/Linux Battery Health Optimization for ASUS Laptops

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1 Introduction

Modern lithium-ion batteries degrade faster when they remain at high charge percentages for long periods of time. For users who continuously use their laptops while plugged into a charger — especially developers, programmers, designers, and office workers — limiting the maximum battery charge level can significantly improve long-term battery health.

Many ASUS laptops support Linux battery charge threshold control directly from the Linux kernel. This allows Ubuntu users to configure battery charging limits without requiring Windows software such as MyASUS.

This documentation explains:

- How battery charge threshold works
- How to configure a permanent 70% charging limit
- How to temporarily enable full charging (100%)
- How to disable the feature completely
- Real-world usage scenarios
- Notes for ASUS and non-ASUS laptops

2 What is Battery Charge Threshold?

Battery charge threshold is a hardware-supported feature that prevents the laptop battery from charging above a specified percentage.

For example:

- If the threshold is set to 70%, the battery will stop charging once it reaches 70%.
- The laptop will then run directly from AC power while plugged in.
- This reduces battery stress and extends battery lifespan.

2.1 Why It Matters

Keeping lithium batteries constantly between 95% and 100% creates heat and voltage stress. Over time, this reduces battery capacity and overall health.

Using a lower threshold:

- Reduces battery wear
- Minimizes heat generation
- Improves long-term battery lifespan
- Keeps battery health stable for years

3 Recommended Threshold Values

Threshold	Recommended Usage
60%	Best for users who keep their laptop plugged in almost all day. Maximum battery lifespan protection.
70%	Balanced option between battery protection and backup battery life. Recommended for developers and office users.
80%	Good for users who occasionally work without a charger for several hours.
100%	Use only when full battery backup is temporarily required for travel or outdoor work.

4 Checking Device Support

Before configuring charge threshold, verify that your laptop supports the feature.

Open Terminal and run:

```
ls /sys/class/power_supply/
```

You should see something similar to:

```
BAT0  
ACO
```

Now check whether the threshold control file exists:

```
ls /sys/class/power_supply/BAT0/
```

If you see:

```
charge_control_end_threshold
```

then your laptop supports battery charge threshold control.

5 Temporary Battery Limit Setup

To temporarily set the battery charge limit to 70%:

```
echo 70 | sudo tee \  
/sys/class/power_supply/BAT0/charge_control_end_threshold
```

This change works immediately but resets after reboot.

6 Permanent Battery Threshold Setup

To make the setting permanent across reboots, create a systemd service.

6.1 Step 1: Create Service File

Run:

```
sudo nano \  
/etc/systemd/system/battery-charge-threshold.service
```

Paste the following configuration:

```
[Unit]  
Description=Set Battery Charge Threshold  
After=multi-user.target  
StartLimitBurst=0  
  
[Service]  
Type=oneshot  
Restart=on-failure  
ExecStart=/bin/bash -c \  
'echo 70 > \  
/sys/class/power_supply/BAT0/charge_control_end_threshold'  
  
[Install]  
WantedBy=multi-user.target
```

Save the file:

- Press CTRL + O
- Press Enter
- Press CTRL + X

6.2 Step 2: Enable the Service

Run:

```
sudo systemctl enable --now \  
battery-charge-threshold.service
```

The 70% threshold will now automatically apply every time Ubuntu starts.

7 How It Works Internally

When the battery reaches 70%:

- Charging stops automatically
- The laptop switches to direct AC power usage
- The battery remains idle
- No forced discharge occurs

Example:

- Current battery level: 92%
- Threshold set to: 70%

The battery will remain at 92% until normal usage gradually reduces it to 70%. After reaching 70%:

- Charging resumes only if the level drops below 70%
- Charging stops again once 70% is reached

8 How to Temporarily Charge to 100%

Sometimes full battery backup is needed for travel, meetings, or outdoor work.

To temporarily enable full charging:

```
echo 100 | sudo tee \
/sys/class/power_supply/BAT0/charge_control_end_threshold
```

Now the battery can charge fully to 100%.

8.1 Returning to 70% Again

Option 1: Reboot the system.

Since the permanent service is already enabled, rebooting automatically restores the 70% limit.

Option 2: Apply manually without reboot:

```
echo 70 | sudo tee \
/sys/class/power_supply/BAT0/charge_control_end_threshold
```

9 How to Disable the Feature Completely

If you no longer want battery threshold control:

9.1 Stop the Service

```
sudo systemctl disable --now \
battery-charge-threshold.service
```

9.2 Delete the Service File

```
sudo rm \
/etc/systemd/system/battery-charge-threshold.service
```

9.3 Reload systemd

```
sudo systemctl daemon-reload
```

9.4 Restore Full Charging

```
echo 100 | sudo tee \  
/sys/class/power_supply/BAT0/charge_control_end_threshold
```

10 Real-World Usage Scenarios

10.1 Scenario 1: Developer / Programmer

- Laptop remains plugged in most of the day
- Heavy coding sessions
- Docker, VS Code, Android Studio, Browsers running continuously

Recommended Threshold:

70%

Benefits:

- Lower battery temperature
- Better battery lifespan
- Reduced long-term degradation

10.2 Scenario 2: Office User

- Mostly desk work
- Occasional unplugged meetings

Recommended Threshold:

70% or 80%

10.3 Scenario 3: Frequent Traveler

- Works outdoors frequently
- Needs long battery backup

Recommended Threshold:

80%

Temporarily switch to 100% before travel.

11 Supported Laptop Brands

11.1 ASUS

Most modern ASUS laptops support this feature natively in Linux.

Examples:

- ASUS TUF Series
- ASUS ROG Series
- ASUS VivoBook
- ASUS ZenBook

11.2 Other Laptop Brands

Some other manufacturers also support Linux battery thresholds:

Brand	Linux Support Status
Lenovo ThinkPad	Excellent Linux support using <code>tp_smapi</code> or native kernel features
Dell	Some models support battery thresholds through BIOS or kernel interfaces
HP	Limited support depending on model
Acer	Varies by hardware generation
MSI	Some gaming models support it

Note: Different manufacturers may use different file paths and control interfaces.

12 Advantages of Battery Threshold Limiting

- Extends battery lifespan
- Reduces heat generation
- Minimizes battery swelling risks
- Improves long-term battery health
- Ideal for always-plugged-in workflows
- No performance reduction
- Works directly from Linux kernel support

13 Important Notes

- This feature does NOT damage the battery.
- The battery is NOT forcibly discharged.
- The laptop safely runs from AC power while plugged in.
- Battery calibration should still occasionally be done.
- Use 100% mode only when necessary.

14 Conclusion

Battery charge threshold management is one of the best practices for maintaining long-term laptop battery health on Linux systems.

For ASUS Ubuntu users, configuring a 70% charging limit provides an excellent balance between:

- Battery protection
- Daily usability
- Long-term performance

Using a permanent systemd service ensures the configuration automatically applies during every system boot without requiring additional software.

This solution is lightweight, kernel-based, efficient, and ideal for developers and professional users who keep their laptops plugged in for extended periods.

References

1. Ubuntu Linux Documentation
2. Linux Kernel Power Supply Subsystem Documentation
3. ASUS Linux Community Documentation
4. systemd Official Documentation
5. Arch Linux Wiki - Laptop Battery Threshold